



Cognitive Mapping of Environmental Risk Perception in New England Using a Scalable Artificial Intelligence Framework

SUBJECT INFORMATION AND INFORMED CONSENT FORM (SBR)

Principal Investigator: Dr. Hanlin Zhou

Student Investigator: Zhang (Ludwig) Chen

Institution: Department of Geography, Sustainability, Community, and Urban Studies, University of Connecticut

Address: Philip E. Austin Building, 215 Glenbrook Rd., Storrs, CT 06269

Telephone: 1 647-879-5378

Sponsor: Institute for the Brain and Cognitive Sciences (IBACS), University of Connecticut

Key Information

The following is a concise and focused presentation of key information to assist you in understanding why you might or might not want to participate in this research.

- You are being asked to take part in a research study because you are an adult ages 18 to 64, you can read English, and you have lived in New England for at least 6 months during the past 12 months. You are seeing this study invitation through Prolific, an online research participant platform.
- Your participation is voluntary. You may refuse to participate or withdraw at any time without penalty or loss of benefits to which you are otherwise entitled. Your alternative is to not take part in the study.
- The purpose of the study is to measure environmental risk perception using an online image comparison task and questionnaires. The results will be used to identify patterns in risk perception and develop research-only analytic models at an aggregated geographic level.
- After the survey, your de-identified image-comparison responses will be converted into image-level aggregated scores and used to train and evaluate an AI-based research scoring model. Your Prolific ID will not be used for model training.

- Your participation in the study is expected to last about 10 minutes and will include 1 online visit.
- The main study procedures include four steps. First, you will confirm your eligibility, review this consent form, and indicate your agreement before any study activities begin. Second, you will complete 20 rounds of image pair comparisons. In each comparison you will see two street images and select an image based on six environmental perception questions. Third, you will complete a short twelve-question questionnaire that includes demographic questions and brief measures related to stress, climate change awareness, and environmental experiences. Fourth, after you complete all study procedures, you will be shown a completion code and redirected back to Prolific for payment processing. You will not need to provide an email address.
- The risks of the study are low and may include potential but limited discomfort from viewing some street view images or answering questions about stress and environmental experiences, and fatigue from completing the online task. We use secure UConn systems and restricted access to reduce privacy risks.
- There is no direct benefit to you from taking part in this study. However, information we learn from the study results may help people in the future.

This overview does not include all the information you need to know before deciding whether to take part. Additional information is given in the rest of this consent form. Be sure to review the rest of this consent form before deciding about participation.

Who is the Study Team and How is the Study Funded?

This study is being conducted by Dr. Hanlin Zhou and Zhang (Ludwig) Chen. The Institute for the Brain and Cognitive Sciences (IBACS), University of Connecticut, has provided funding for this study.

How Many People Will Take Part in the Study?

The online study will include approximately 1,550 study participants from New England.

What is the Purpose of this Study?

The purpose of this study is to measure environmental risk perception using an online image comparison task and questionnaires. The results will be used to identify patterns in risk perception and develop research-only analytic models at an aggregated geographic level.

What is Involved in this Research Study?

If you agree to be in this study, your participation will involve confirming your eligibility, reviewing the consent form, finishing 20 rounds of image comparison, and completing a short questionnaire. You will not interact with AI or AI-related contents. No chatbot, adaptive survey, automated feedback, or AI-mediated communication will be used. Your participation will last for about 10 minutes in a single online session. Below are the details:

- Confirm eligibility, including age, English ability, and New England residency.
- Review this consent form and indicate your agreement before any study activities begin.
- Complete 20 rounds of image pair comparisons. In each comparison you will see two street images and select an image based on six environmental perception questions.
- Complete a short questionnaire that includes 12 questions. It contains demographic questions and brief measures related to stress, climate change awareness, and environmental experiences.
- After you complete all study procedures, you will be shown a completion code and redirected back to Prolific for payment processing. You will not need to provide an email address.

Will Participating in this Study Benefit Me?

There is no direct benefit to you by participating in this study.

What Are the Risks Associated with This Research?

The risks involved with participation in this study are low and may include potential but limited discomfort from viewing some street-level images or answering questions about stress and environmental experiences, and fatigue from completing the online task. We use secure UConn systems and restricted access to reduce privacy risks. AI-related privacy risk is also minimal because model development will use de-identified aggregated perception scores and street view image features, and your Prolific ID will not be included in model inputs or outputs.

Do I Have to Participate?



No, your participation is voluntary. You may decide not to participate at all or, if you start the study, you may withdraw at any time. Withdrawal or refusing to participate will not affect your relationship with the University of Connecticut in any way.

If you would like to participate, click the 'I agree' button and proceed to the online study. You may retain a copy of this consent form by using your browser's Print function to print this page before starting the survey.

What Are the Costs of Taking Part in This Study?

There are no costs to you for taking part in this study.

Will Be Paid for Participating?

Yes, you will be paid through Prolific's built-in payment system upon verified completion of the entire online study, including all 20 image comparisons and the questionnaire. Compensation will follow Prolific's fair payment guidelines (\$2 for 10 minutes). You will receive compensation within 5 business days after you finish the study, provided that you submit complete and valid responses. The research team will not collect your email address or process payment directly. We will not collect your Social Security Number. Prolific's standard completion verification and payment policies will apply. There is no partial compensation for incomplete participation.

If you do not wish to be paid for your participation in this study, please initial here: _____

What About Confidentiality?

We will make every effort to protect the confidentiality of study information that identifies you, but we cannot guarantee total confidentiality. Survey responses will be stored without direct identifiers and accessed only by the PI and the student investigator. No personal identifiable information will be collected by the research team. The only participant-level identifier recorded will be your Prolific ID, used solely to verify completion and process payment. Prolific IDs are stored separately from research responses, are not linked to survey data, are accessible only to the PI, and will be deleted after payment reconciliation is complete. All data will be stored on UConn-managed systems, including UConn OneDrive storage and the PI's institutional desktop computer. Results will be reported only in aggregate, and no identifying information will be used in publications or presentations.



Data will be collected using the Internet; no guarantees can be made regarding the interception of data sent via the Internet by any third party. Confidentiality will be maintained to the degree permitted by the technology used.

Who Do I Call If I Have Questions or Problems?

If you have any questions about this study, please contact Dr. Hanlin Zhou (Phone: +1 647-879-5378; Email: hanlin.zhou@uconn.edu). If you have any questions or complaints, you may contact a person not on the research team at the Biomedical Research Alliance of New York Institutional Review Board at (516) 318-6877 or at www.branyirb.com/concerns-about-research.

Consent Agreement:

If you click the 'I agree' button below, it means that you read this consent form and agreed to participate in this study. You may retain a copy of this consent form by using your browser's Print function to print this page before starting the survey.

(Buttons):

I agree

I refuse